

FINAL CROSS-VENUE RWA BASIS TRADE ANALYSIS

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STRATEGY DEFINITION

$\text{basis_bps} = 10,000 * (\text{Hyperliquid_mid} / \text{Lighter_mid} - 1)$

Positive basis: Hyperliquid rich, short Hyperliquid / long Lighter.

Negative basis: Lighter rich, long Hyperliquid / short Lighter.

This is cross-venue basis convergence, not classic two-asset pair trading.

DATA SET

Assets: BRENT OIL, GOLD, SILVER, WTI

Fresh matched seconds: 92,079

Date range: 2026-04-23 08:17:56+00:00 to 2026-04-25 07:49:56+00:00

Freshness rule: both venues within 2 seconds of exchange timestamp.

Base size: \$10,000 per leg, \$20,000 gross notional per single-entry trade.

BASIS RICHNESS

BRENT OIL mean= 1.58 bps, p90_abs= 3.44 bps, rich venue=Hyperliquid, crosses=1520

GOLD mean= 1.91 bps, p90_abs= 3.63 bps, rich venue=Hyperliquid, crosses= 238

SILVER mean= -2.37 bps, p90_abs= 5.73 bps, rich venue=Lighter , crosses=1087

WTI mean= -2.36 bps, p90_abs= 5.17 bps, rich venue=Lighter , crosses=1160

NO-MAX-HOLD MID-PRICE RESULTS

Single-entry trades: 4,224

Single-entry total PnL: \$5,717.90

Single-entry win rate: 84.23%

Single-entry convergence/cross rate: 99.91%

Single-entry avg hold: 104.55s

Scaled total PnL, 1 bps step / 5 layer cap: \$13,667.57

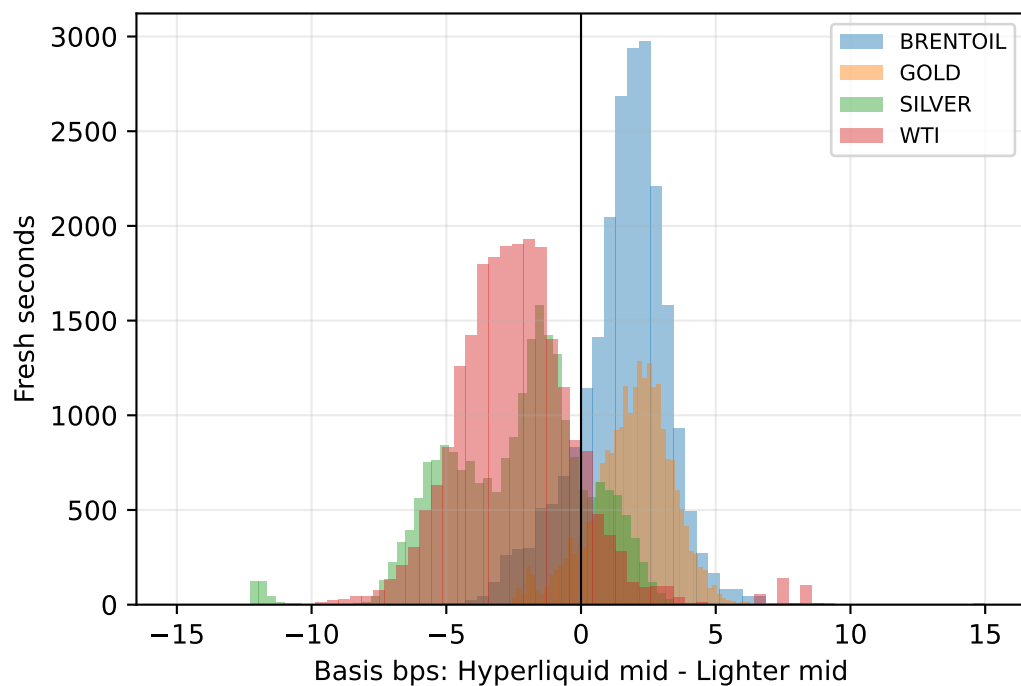
Scaled avg layers: 1.43

Scaled avg max notional per leg: \$14,259.00

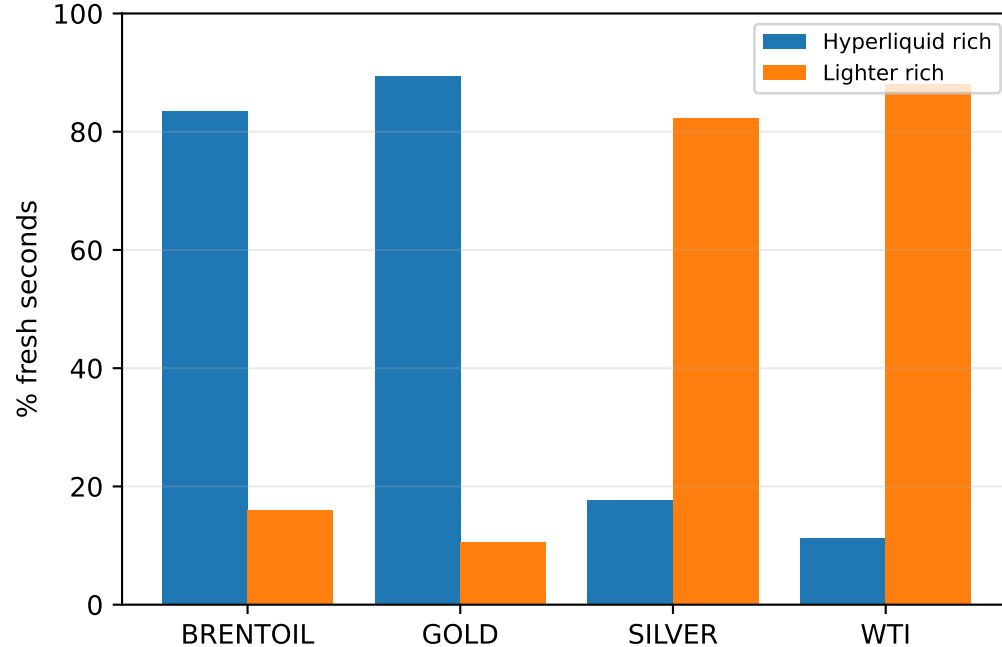
Best tested scale: 0.5 bps step, 10 layers, PnL \$17,034.64

Basis Distribution and Venue Richness

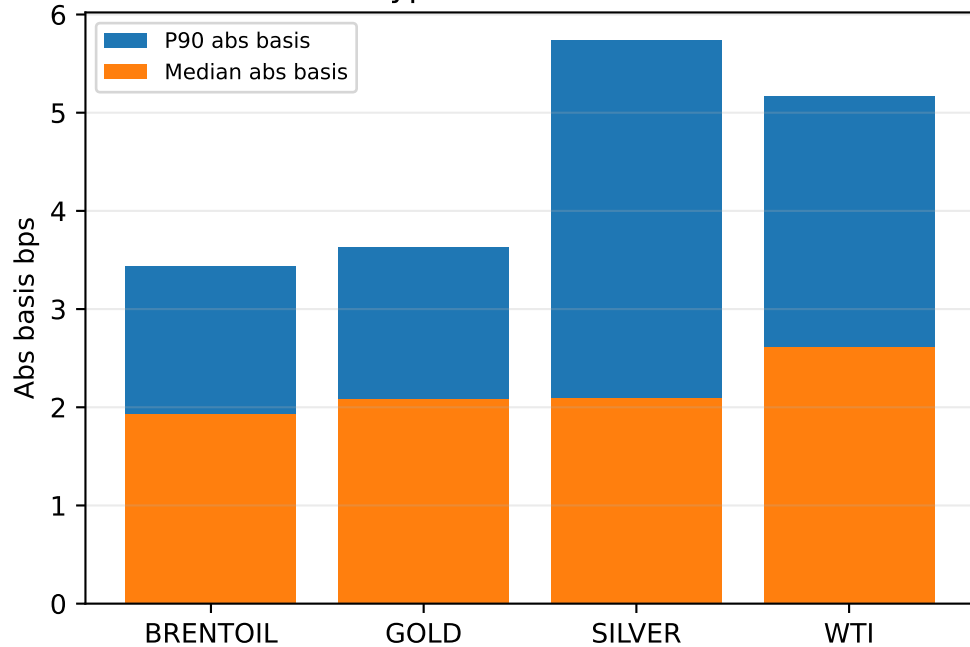
Basis Distribution



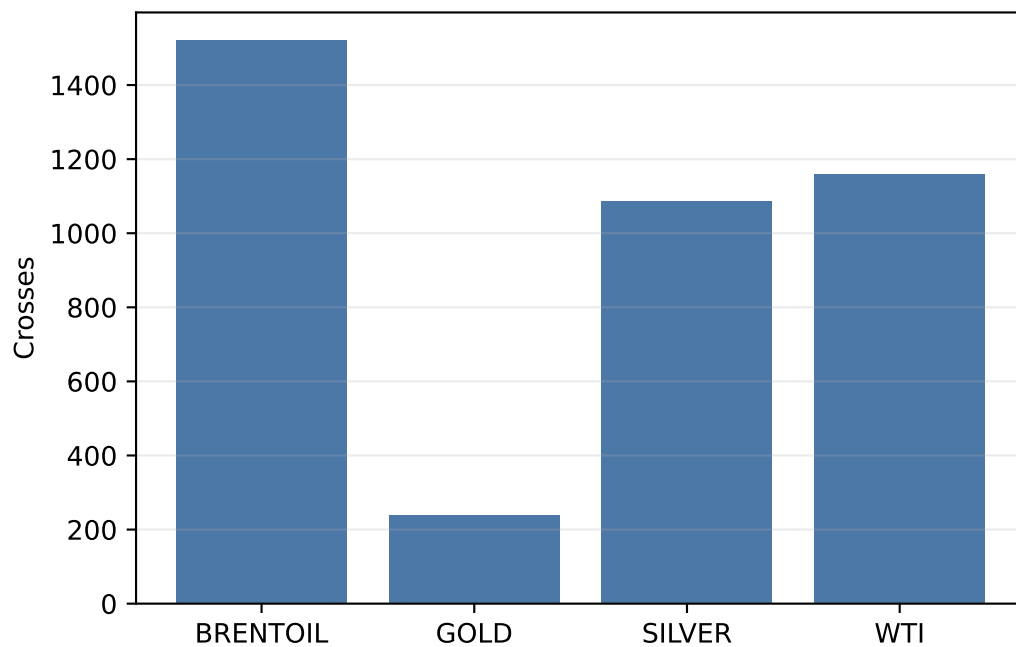
Dominant Rich Venue



Typical vs Tail Basis

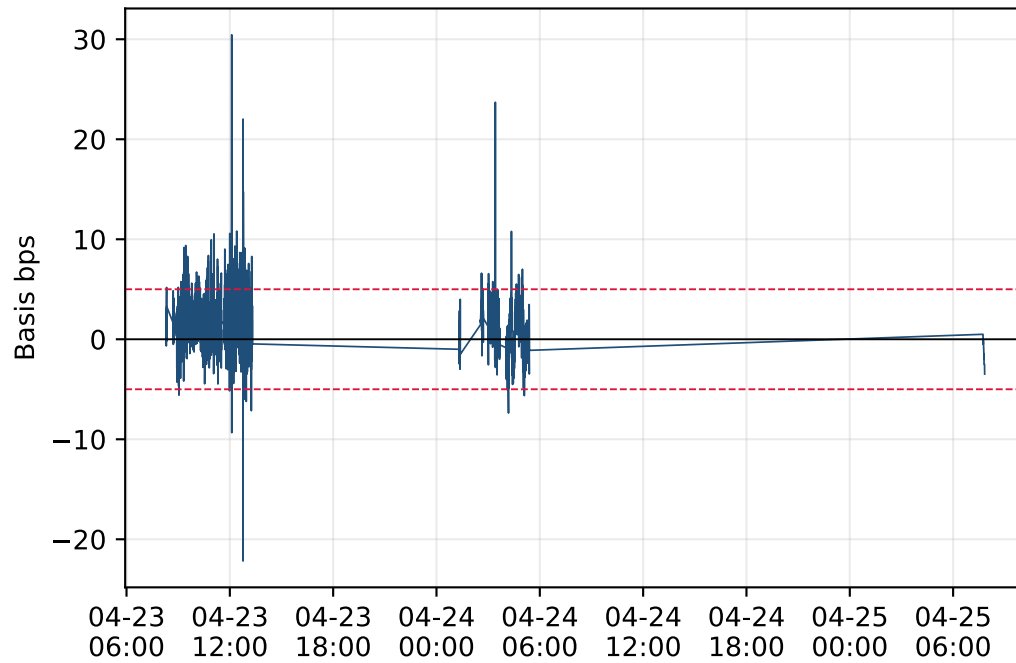


Basis Zero-Cross Count

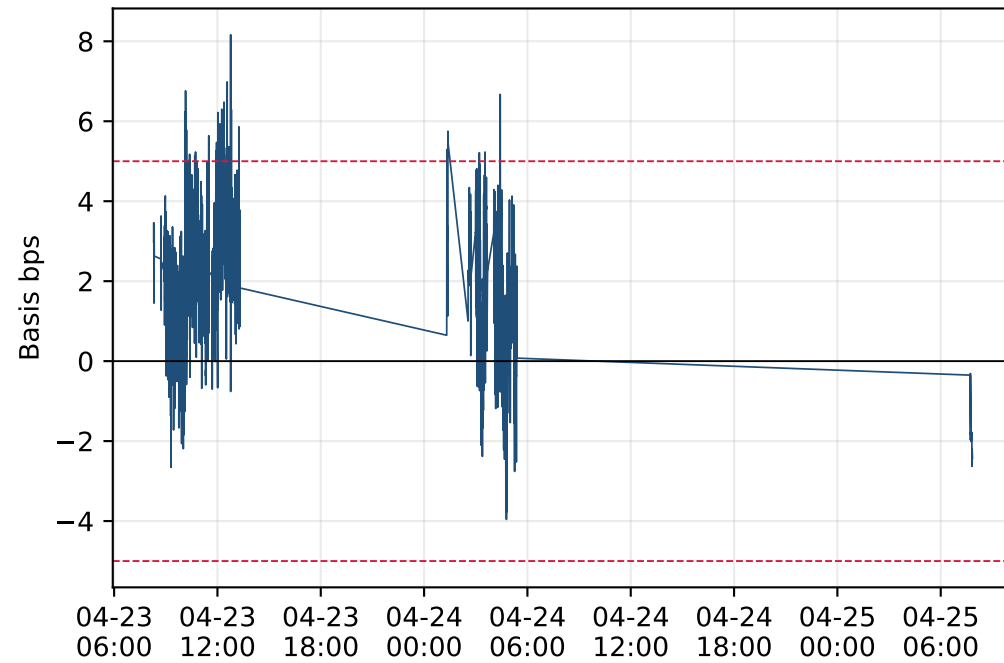


Basis Timing

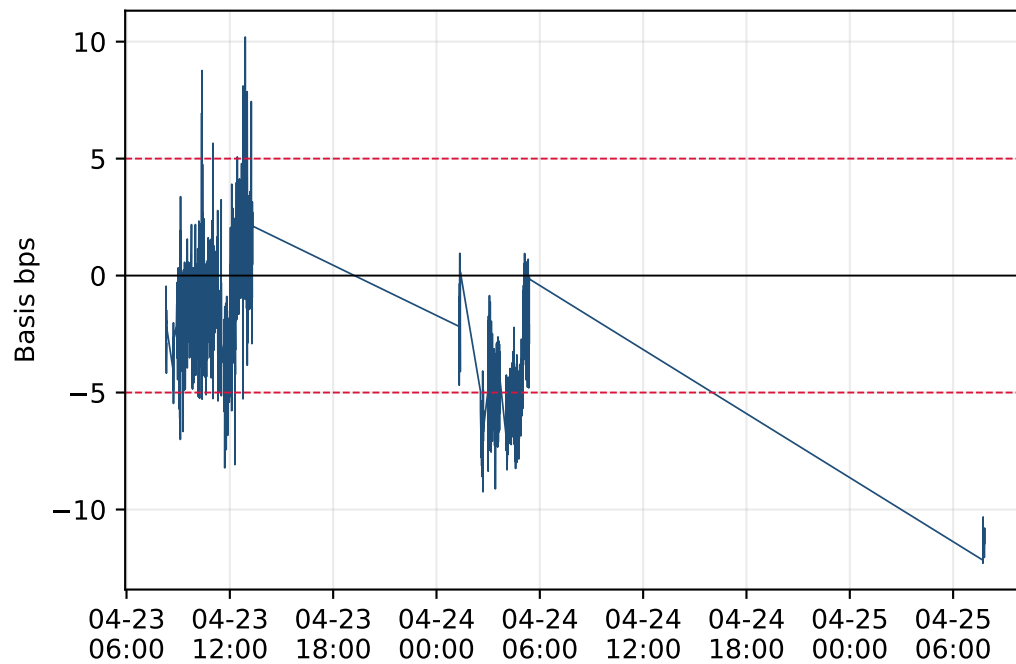
BRENTOIL



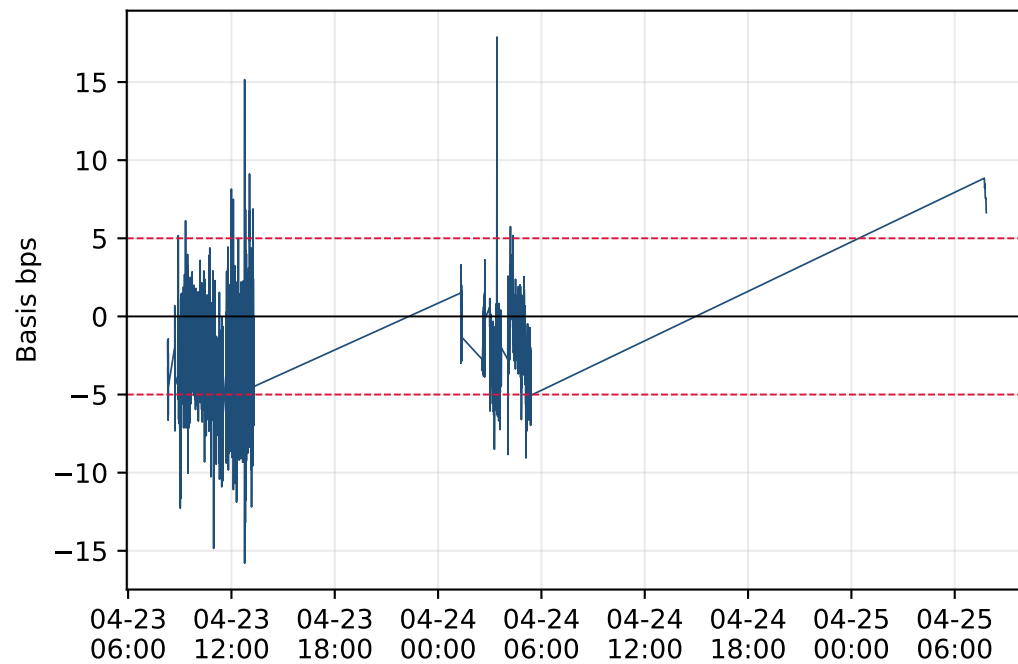
GOLD



SILVER

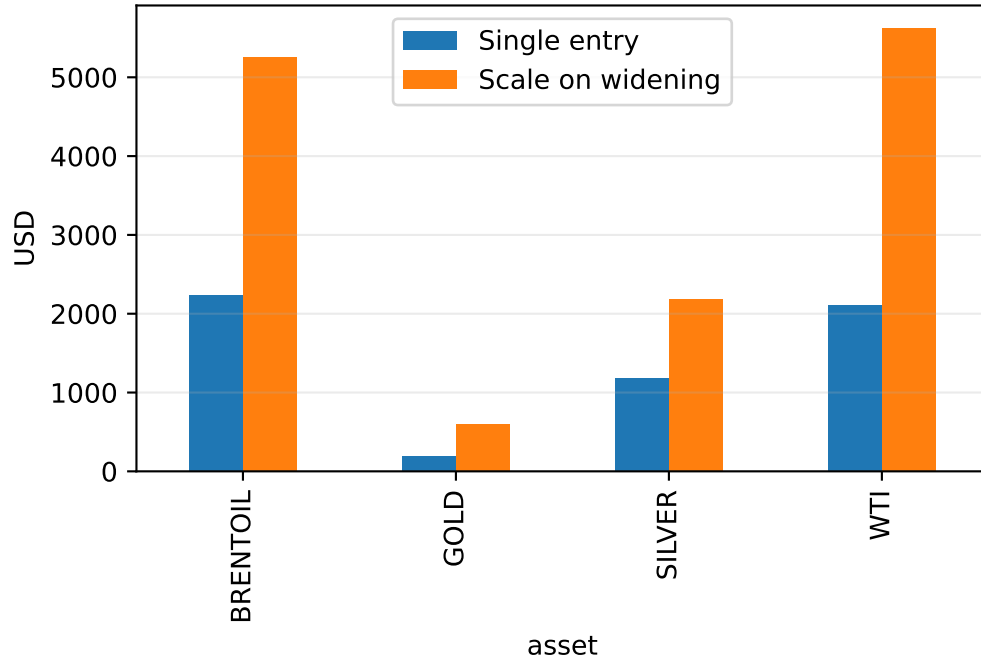


WTI

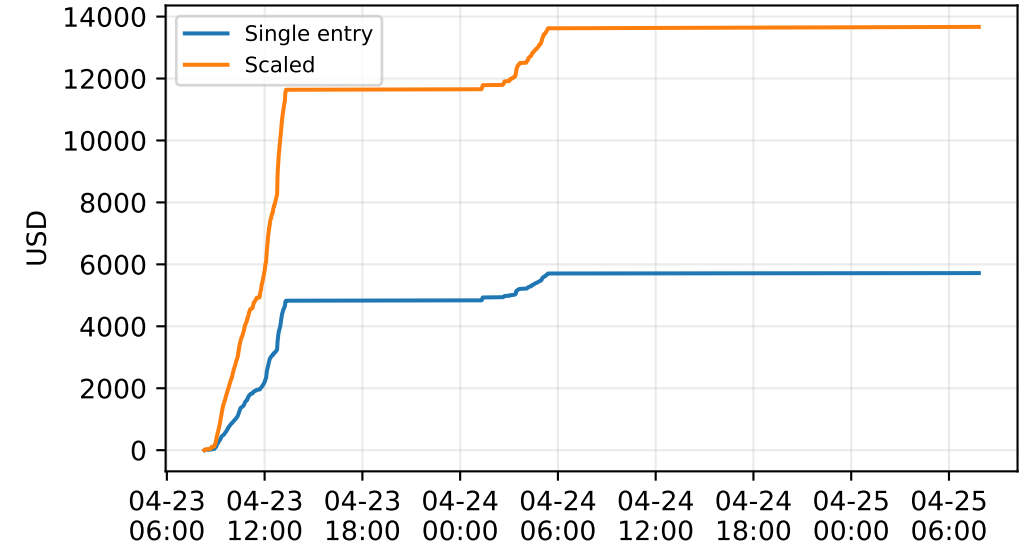


Basis Trade PnL and Scaling

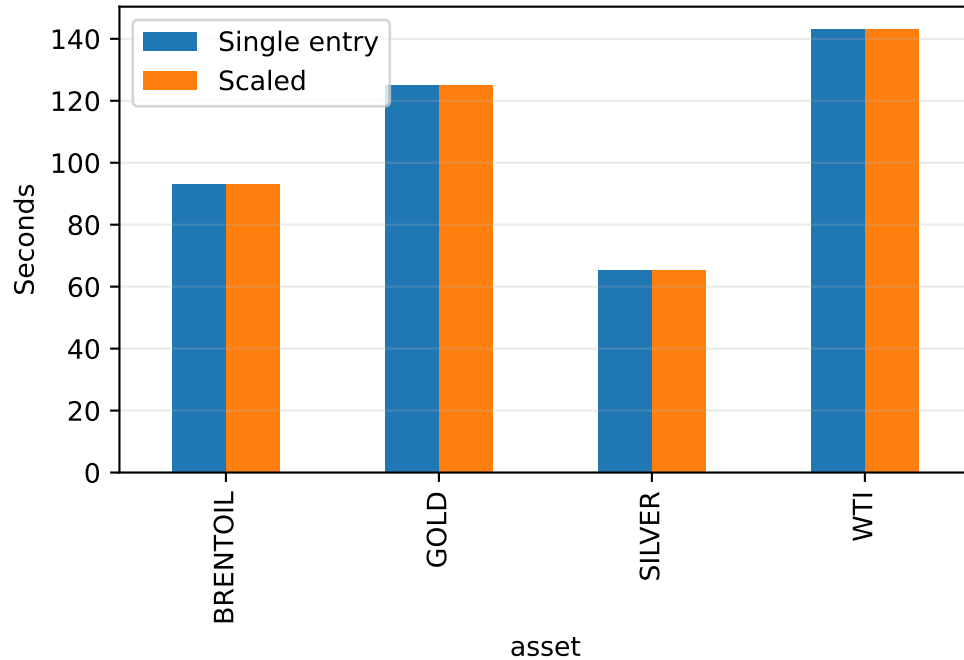
PnL by Asset



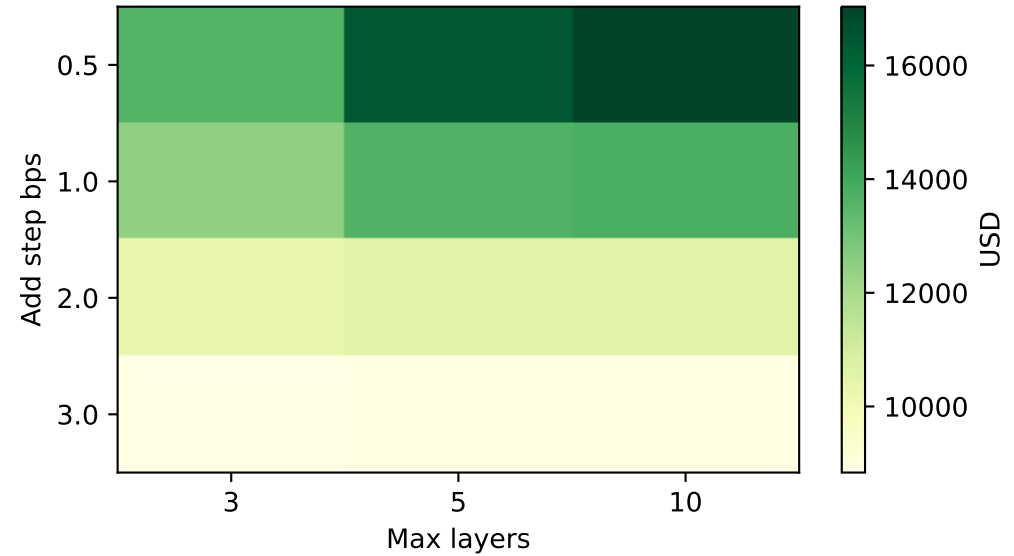
Cumulative Mid-Price PnL



Average Hold

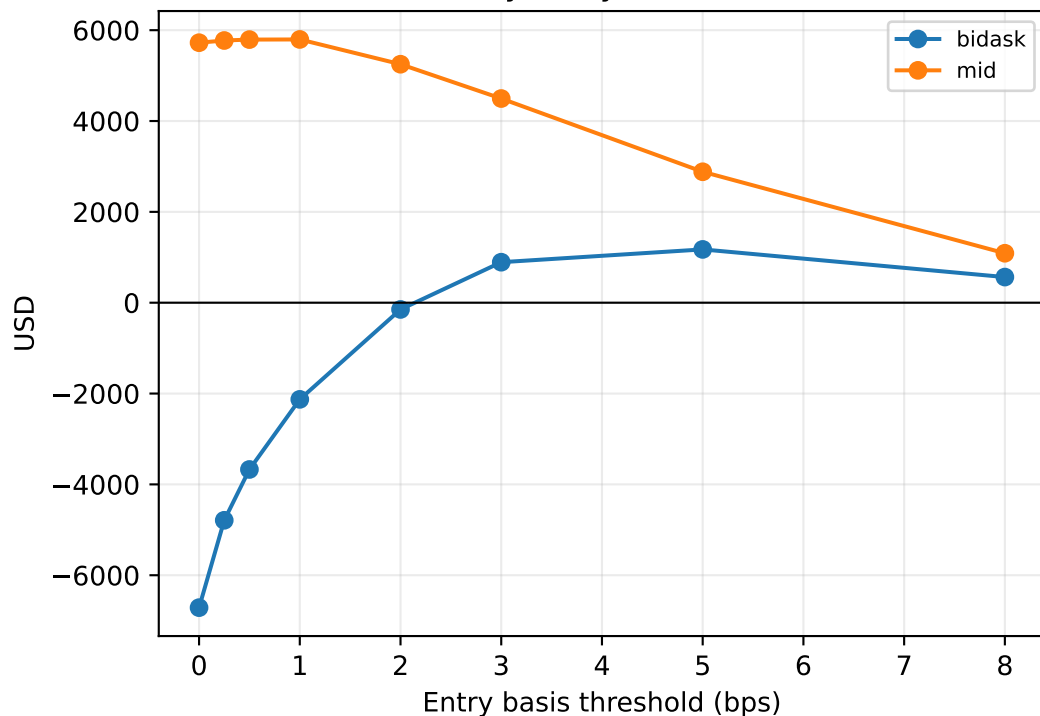


Scaling Sensitivity PnL

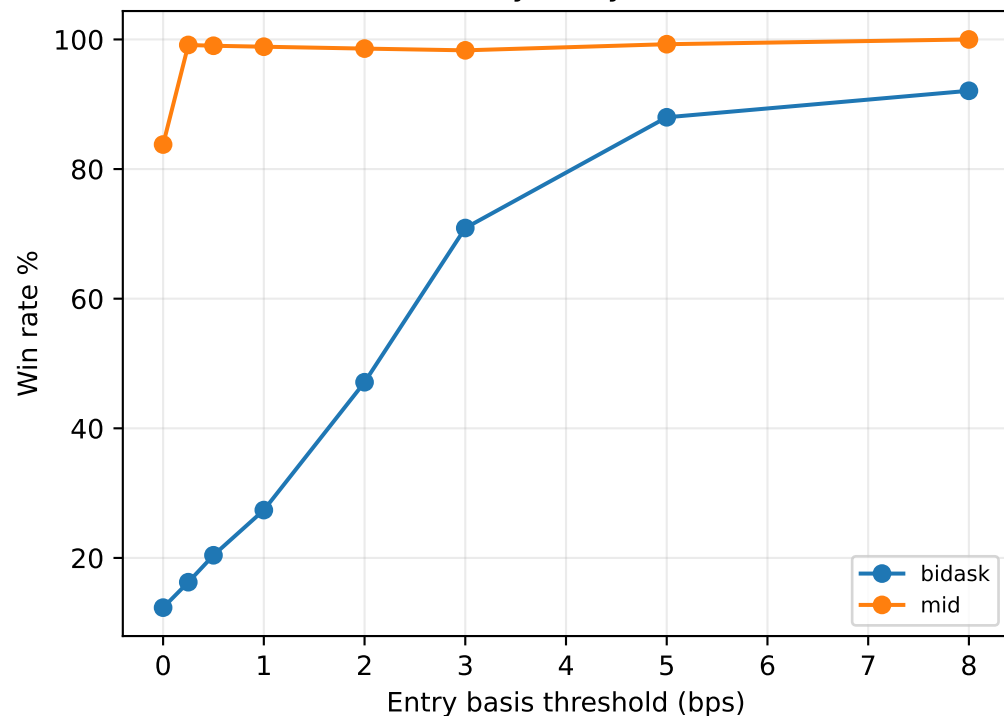


Execution Reality: Mid Alpha vs Bid/Ask Crossing

PnL by Entry Threshold



Win Rate by Entry Threshold



STRATEGY READOUT

SINGLE-ENTRY MID-PRICE PNL BY ASSET

BRENT OIL	trades=1356,	pnl= \$2,237.20,	win= 91.0%,	avg_hold= 93.2s
GOLD	trades= 606,	pnl= \$192.70,	win= 53.3%,	avg_hold= 125.0s
SILVER	trades=1083,	pnl= \$1,178.04,	win= 92.7%,	avg_hold= 65.2s
WTI	trades=1179,	pnl= \$2,109.96,	win= 84.6%,	avg_hold= 143.2s

SCALED MID-PRICE PNL BY ASSET

BRENT OIL	trades=1356,	pnl= \$5,252.98,	win= 91.4%,	avg_layers=1.49,	avg_leg_notional= \$14,859.88
GOLD	trades= 606,	pnl= \$598.45,	win= 54.0%,	avg_layers=1.23,	avg_leg_notional= \$12,277.23
SILVER	trades=1083,	pnl= \$2,186.41,	win= 94.1%,	avg_layers=1.30,	avg_leg_notional= \$13,047.09
WTI	trades=1179,	pnl= \$5,629.72,	win= 85.0%,	avg_layers=1.57,	avg_leg_notional= \$15,699.75

EXECUTION REALITY CHECK

Ideal mid fills show the alpha process. Crossing every tiny gap does not.

Bid/ask entry gap > 0.00 bps:	trades=4274,	pnl= \$-6,713.65,	win= 12.3%
Bid/ask entry gap > 0.25 bps:	trades=3297,	pnl= \$-4,790.40,	win= 16.3%
Bid/ask entry gap > 0.50 bps:	trades=2876,	pnl= \$-3,671.41,	win= 20.4%
Bid/ask entry gap > 1.00 bps:	trades=2319,	pnl= \$-2,127.96,	win= 27.4%
Bid/ask entry gap > 2.00 bps:	trades=1488,	pnl= \$-148.40,	win= 47.1%
Bid/ask entry gap > 3.00 bps:	trades= 952,	pnl= \$891.92,	win= 70.9%
Bid/ask entry gap > 5.00 bps:	trades= 408,	pnl= \$1,173.33,	win= 88.0%
Bid/ask entry gap > 8.00 bps:	trades= 101,	pnl= \$564.45,	win= 92.1%

FINAL VERDICT

The basis is highly mean-reverting in the sample and supports a real alpha thesis.

The strategy should be described as cross-venue RWA basis convergence.

Scaling into widening improved ideal PnL, but it increases tail exposure.

Production must use maker-biased execution, freshness checks, funding-rate monitoring, and layer caps.